sent to the brain via the auditory nerve and the brain interrupts them as sound.

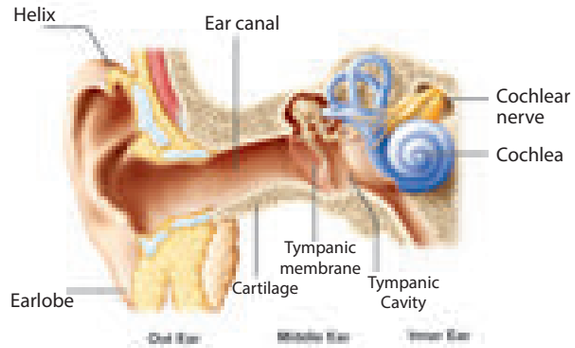


Figure 8.10 Human ear

Points to Remember

- Sound is produced due to vibration.
- Sound cannot travel through vacuum.
- Sound travels as a longitudinal wave through a material medium.
- Sound travels as successive compressions and rarefactions in the medium.
- In sound propagation, it is the energy of the sound that travels and not the particles of the medium.
- The speed (v), frequency (n), and wavelength (λ), of sound are related by the equation, $v = n \lambda$.
- The law of reflection of sound: (i) The angle of incidence, the angle of reflection and normal drawn at the point of incidence all lie in the same plane (ii) The angle of incidence (i) and the angle of reflection (r) are always equal.
- The speed of sound depends primarily on the nature and the temperature of the transmitting medium.
- For hearing distinct echo sound, the time interval between the original sound and the reflected sound must be at least 0.1 s.
- The persistence of hearing sound in an auditorium is the result of repeated reflections of sound and is called reverberation.
- The amount of sound energy passing each second through unit area is called the intensity of sound.
- The audible range of hearing for average human being is in the frequency range of 20 Hz to 20000 Hz
- Sound waves with frequencies below audible range are termed as 'Infrasonics' and those above audible range are termed as 'Ultrasonics'.
- The SONAR technique is used to determine the depth of the sea and to locate under water hills, valleys, submarines, icebergs, etc.

A-Z GLOSSARY

Amplitude	The maximum displacement of a particle.
Compressions	The region of increased pressure.
Echo	The repetition of sound caused by the reflection of sound.
Frequency	Number of waves produced in one second.
Longitudinal wave	The wave that propagates with compressions and rarefactions.
Pitch	Characteristics of sound based on frequency.
Rarefactions	The region of decreased pressure.
Reverberation	The repeated reflection that results in persistence of sound is called reverberation.
Timbre (or quality)	Characteristic which distinguishes the two sounds of same loudness and pitch emitted by two different instruments.
Time period	Time taken to produce one wave.
Ultrasonic sound	Sound waves with frequencies greater than 20,000 Hz.
Wave	The propagating disturbance that travels in a medium.
Wavelength	The minimum distance in which a sound wave repeats itself.

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